Mahmoud Alyosify

Computer Vision Engineer

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SUMMARY

Computer vision engineer focused on image models that **learn with few labels and are ready to deploy**. Built a self-supervised image model (SimCLR) that reaches 84.30% top-1 accuracy and serves image search through ONNX and FAISS, and led a system that reads vital signs from ordinary video (Microsoft Imagine Cup 2023 semi-finalist). MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30), with graduate coursework and certification in deep learning.

SKILLS

Computer vision: CNNs, image classification, self-supervised and contrastive learning (SimCLR, SupCon), image retrieval, data augmentation

Video & signals: rPPG, video signal processing, spectrograms

Tools: PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Matplotlib

Deployment: ONNX, FAISS, FastAPI, Streamlit, Docker, AWS

Programming: Python, C++, SQL, Git, Linux

PROJECTS

SimCLR-Vision-SSL — self-supervised image model

2026

PyTorch, SimCLR, SupCon, ONNX, FAISS

- Trained an image model without labels (SimCLR), tested **42 augmentation settings** and reached **84.30% top-1 accuracy** with a semi-supervised extension (SupCon).
- Ran a controlled 42-experiment study to find which image augmentations actually improve the model; best result 84.30% top-1 accuracy.
- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

Vitalism Solution — graduation project

2023

Python, computer vision, signal processing (rPPG)

- Measures heart rate and other vital signs from ordinary video, with no contact sensors. Grade **A+**; **Microsoft Imagine Cup 2023 semi-finalist**.
- Led a six-student team from idea to a working prototype; Finalist at ISEIC 2023.

AudioShield — deepfake audio detection

Python, CNN, Streamlit

- Trained a CNN on spectrograms to tell real speech from synthetic speech; served it in a Streamlit app with live predictions and PDF reports.

SCRAP — satellite collision risk prediction

2026

Python, scikit-learn, imbalanced data

- Predicted satellite collision risk using only data available **48+ hours** before closest approach, so the warning arrives in time to act.

Also: MinimalLM (MSc thesis, 47.6% fewer LLM tokens) · Horus-OSINT (Llama-3-8B fine-tuned on AWS) · HORUS Sentinel (multi-agent OSINT).

EXPERIENCE

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group) — Maadi, Cairo

- Built a RAG assistant that answers staff questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut false refusals from 5 in 20 to 1 in 20 through measured design changes.

Part-Time Machine Learning Instructor — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.

AI in Cybersecurity Intern Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) Feb 2024 – Mar 2025
National Company for Roads Building & Development Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.
- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and Crystal Reports.

Online Instructor 2020 – Present
Udemy & YouTube (Arabic courses) Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS

- Deep Learning Specialization (DeepLearning.AI, 2026)
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- IBM AI Developer Professional Certificate (IBM, 2026)
- AWS Academy: Machine Learning Foundations, ML for NLP, Data Engineering
- HCIP-AI (Huawei)
- Machine Learning Internship, ITIDA / Egypt Makes Electronics (200 hours)

AWARDS & LEADERSHIP

- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)
- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Ideal Student Award, Faculty of Computers & Information, Assiut University (2022–2023)

LANGUAGES

Arabic (native) · English (professional working proficiency)